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Optimal portfolio strategies for a large universe of assets

In this project we are interested in portfolio strategies that work well for large universes of assets. The starting point for optimal portfolio strategies is the Markowitz' mean-variance portfolio, with two assumptions: (i) the future return can be predicted using lagged regressors; and (ii) the covariance matrix of returns is constant.

Two issues appear in this framework. As concerns (i), the out-of-sample R^2 of the best predictors is only approximately 1%; and as concerns (ii), there are too many elements in the covariance matrix to estimate it precisely. In high-dimensional settings, where we have many more assets than the sample size, these problems are exacerbated.

Let's be precise. Consider a vector of N returns $r_t = (r_{1t}, \dots, r_{Nt})'$, with N potentially larger than the sample size T . Conditional on all past information, its distribution is as follows

$$r_t \sim N(\mu_{t|t-1}, \Sigma), \text{ for all } t \geq 2.$$

The entries of the predictive mean vector $\mu_{t|t-1}$ will be obtained from predictive regressions (Welch & Goyal, 2008) and could be altered by economic restrictions (Campbell & Thompson, 2008) or shrinkage methods (see, for example, DeMiguel et al., 2009). More modern approaches exist, such as Gu et al. (2020) and Kelly & Malamud (2023).

Many new advances are made in the estimation of the covariance matrix Σ . A very competitive one is the nonlinear shrinkage method of Ledoit & Wolf (2020), but many competitors exist (see the references therein).

Can we create optimal combinations of models for (i) and (ii) in terms of common performance measures for portfolio returns, like the Sharpe ratio, certainty equivalent return, and portfolio variance?

References

Skim the first three papers for a general understanding of the approaches in the literature. The other papers provide some interesting directions for research.

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- 2) DeMiguel, V., Garlappi, L., and Uppal, R. (2009). Optimal versus naive diversification: How inefficient is the 1/n portfolio strategy? *The Review of Financial Studies*, 22(5):1915–1953.*
- 3) Ledoit, O. and Wolf, M. (2020). Analytical Nonlinear Shrinkage of Large-Dimensional Covariance Matrices. *The Annals of Statistics*, 48(5):3043–3065.*
- 4) Campbell, J. Y. and Thompson, S. B. (2008). Predicting excess stock returns out of sample: Can anything beat the historical average? *The Review of Financial Studies*, 21(4):1509–1531.
- 5) Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273.
- 6) Kelly, B. T., Malamud, S., and Zhou, K. (2023). The virtue of complexity in return prediction. *The Journal of Finance*. Forthcoming.

Forecasting Realized Volatility Using the VIX in a Nonlinear Framework

Supervisor: Jeroen Dalderop

The availability of high-frequency return data allows accurately estimating the volatility during a period (such as day or month) by the realized volatility measure $RV_t = \sum_{i=1}^M r_{t,i}^2$, where $r_{t,i}$ are intra-period returns. Forecasts of realized volatility are a key input for applications in portfolio allocation, risk management, and derivatives pricing. The CBOE Volatility Index (VIX) provides a forward-looking measure of realized stock market volatility derived from option prices. While the VIX contains information about future realized volatility, it also contains a variance risk premium when investors are averse to unexpected increases in volatility.

Bekaert and Hoerova (2014) decompose the squared VIX into expected variance and a variance risk premium by linear regressions of RV_{t+1} on VIX_t and other predictor variables X_t such as RV_t . However, more recent models suggest that the relationship between the realized volatility and the VIX is nonlinear and varies across market conditions. Such models emphasize the importance of changing regimes (calm versus crisis times) (Rombouts et al., 2020), leverage effects (Hansen et al., 2024), and/or time-varying volatility risk aversion (Hansen and Tong, 2026).

A general nonlinear model involves covariate-dependent coefficients:

$$RV_{t+1} = \alpha(X_t) + \beta(X_t)VIX_t + \varepsilon_{t+1},$$

The time-varying coefficients flexibly model the state-dependent relation between the VIX and realized volatility. Key modelling choices are which covariates X_t to include, and how to model the coefficient functions $\alpha(\cdot)$ and $\beta(\cdot)$. The latter can either be obtained from a parametric asset pricing model, or estimated unstructured using nonparametric / machine learning methods.

Potential research questions are

- Does allowing for a nonlinear and/or state-dependent impact of the VIX improve realized volatility forecasts?
- Under which market conditions is the VIX most informative for realized volatility, and why?
- Does the nonlinearly estimated variance risk premium predict stock returns?

References

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- HANSEN, P. R. AND C. TONG (2026): “Option pricing with time-varying volatility risk aversion,” *The Review of Financial Studies*, 39, 875–924.
- ROMBOUTS, J. V., L. STENTOFT, AND F. VIOLANTE (2020): “Dynamics of variance risk premia: A new model for disentangling the price of risk,” *Journal of Econometrics*, 217, 312–334.

Robust Score-driven Time Series Models

Supervisor: Ramon de Punder

Breakthroughs in academic research not infrequently arise from combining perspectives of different disciplines. A famous example is the heat equation solution (from physics) to the Black-Scholes model's differential equation (from financial econometrics). Likewise, financial econometricians can learn a lot from machine learners, and vice versa. A typical example is model averaging, where, as opposed to traditional econometric model selection, a weighted average of a collection of models is used for prediction. Another example is regularisation, such as Ridge and LASSO regression, very explicitly trading off the model-implied bias and the variance. For complex models, regularisation is crucial, as such models will otherwise be prone to overfitting.

In this research, you are invited to adapt the original Generalized Autoregressive Score (GAS) model developed by Creal, Koopman, and Lucas (2013) and Harvey (2013), inspired by ideas from machine learning. The main task is to robustify the update step in this explicit gradient updating step; see Lange, Van Os, and Van Dijk (2022) for the classification of updating schemes into explicit and implicit gradient methods. Robustification can be achieved in many different ways, for instance, by adding some adaptive penalty term or adopting a localized likelihood approach (see e.g. Diks, Panchenko, and Van Dijk, 2011).

A more recent paper by Blasques, Francq, and Laurent (2023) suggests that updating by something other than the gradient of the model log-likelihood can be allowed for from the perspective of the dynamic properties of the filter. The modern GAS literature hence allows for a lot of creativity, and you can choose yourself whether you want to embrace that from a more theoretical or empirical perspective in your thesis.

References

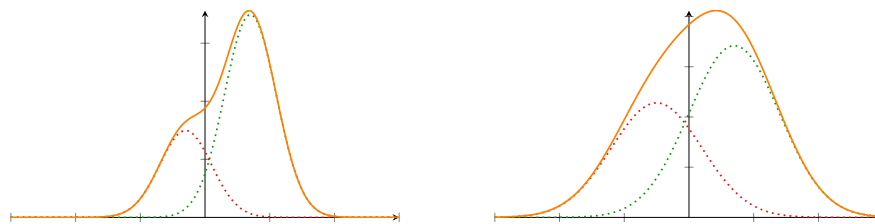
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Modeling Cancer Heterogeneity using Gaussian Mixture Models

Thesis supervisor: Jeroen Eijkens

Description of project: There are different types of cancer, which are usually classified by the part of the body where the tumor first occurred. However, from the location tumor itself it is usually not visible what the cause of the tumor was. Cancer is (largely) a disease of genomic origin, and the differences in genetic mutations that cause tumors in the same body part may be large. Most genes show normal expression across all samples, but some genes will show aberrant expression in parts of the sample.

The aim of this project is to find a mathematical model that enables detecting such multi-modal genes: genes that express differently in one part of samples than in the rest (see the figure below). Such genes may give leads to how the cancer has developed differently across individuals. This in turn may have consequences for the optimal treatment of patients.



Research objective: The main goal of this project is to identify and characterize heterogeneity in cancer gene expression data using Gaussian mixture models, and to classify patients into subgroups. More specifically, the project aims to:

1. Detect genes whose expression is better described by a mixture of distributions rather than a single Gaussian.
2. Implement a suitable (expectation maximization) algorithm to maximize the likelihood of mixture model and estimate the mixture model parameters.
3. Use model selection criteria to decide whether a 2-component model outperforms a 1-component model.
4. Assign patients to latent subgroups via posterior (membership) probabilities and explore clustering of patients based on these probabilities.

You will receive VUmc gene expression data from head-and-neck tumors. We will focus on a limited number of genes (about 150), that are known to correlate significantly to either the patient's survival or the tumor stage.

Useful reading:

- Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006. Chapter 9: Mixture Models and EM.
- Jin Hwan Do, Dong-Kug Choi, *Clustering Approaches to Identifying Gene Expression Patterns from DNA Microarray Data*, Molecules and Cells, Volume 25, Issue 2, 2008, [https://doi.org/10.1016/S1016-8478\(23\)17582-0](https://doi.org/10.1016/S1016-8478(23)17582-0)
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The impact of migration flows on labour market outcomes: evidence from Dutch municipalities and neighbourhoods

Supervisor: Dr. Tommaso Giommoni

Migration plays a central role in shaping labour market outcomes, affecting both employment and wages of native workers. A large and growing literature studies these effects, highlighting that immigration acts as a labour supply shock with potentially heterogeneous impacts across skill groups (see Edo, 2018 for a review). While many studies find small average effects on native wages and employment, the distributional consequences can be substantial, depending on the degree of substitutability between natives and migrants and on labour market institutions. Empirical evidence remains mixed, with some studies documenting negative effects for competing workers and others finding neutral or even positive impacts (e.g. Zorlu, Hartog, 2003; Glitz, 2012; Basso, Peri, 2015). A key challenge in this literature is the endogenous location choice of migrants, which has led to the widespread use of instrumental variable strategies (Jaeger et al., 2018).

This project aims to analyze the impact of migration flows—both international and internal—on labour market outcomes in the Netherlands, focusing on employment and wages of the native population. Adopting a local and dynamic perspective, the analysis will exploit variation across Dutch municipalities and neighbourhoods to study how changes in population composition affect local labour markets. The Netherlands provides an ideal setting due to the availability of detailed and high-frequency administrative data.

Students will rely on publicly available data from Statistics Netherlands (CBS), which provides detailed information on migration, employment, wages, and demographic characteristics at a highly disaggregated geographical level. The empirical analysis will employ micro-econometric techniques, ranging from standard OLS regressions to causal identification strategies based on instrumental variables, such as the commonly used shift-share approach (Jaeger et al., 2018).

The richness of the data allows for several extensions. In addition to labour market outcomes, students may explore how migration affects other local economic dimensions, including housing markets, education, health outcomes, crime, and land use patterns, providing a broader understanding of the economic and social impact of migration.

Data source:

*Data on migration and economic outcomes at the municipal level

https://opendata.cbs.nl/portal.html?_la=nl&_catalog=CBS&tableId=70072ned&_theme=244

*Data on migration and economic outcomes at the neighborhood level

https://opendata.cbs.nl/portal.html?_la=nl&_catalog=CBS&tableId=84286NED&_theme=244

References:

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*Glitz, A. (2012). The labor market impact of immigration: A quasi-experiment exploiting immigrant location rules in Germany. *Journal of Labor Economics*, 30(1), 175-213.

*Zorlu, A., & Hartog, J. (2005). The effect of immigration on wages in three European countries. *Journal of population economics*, 18(1), 113-151.

The local impact of social housing on house market, inequality and labour market outcomes

Supervisor: Dr. Tommaso Giommoni

Social housing policies represent a key example of place-based interventions aimed at improving housing affordability and reducing inequality. Governments allocate substantial resources to these programs, which can significantly reshape urban environments and affect both market outcomes and individual economic trajectories. A growing literature shows that social housing can influence local housing prices and rents (Koster and Van Ommeren, 2019; Li et al., 2025), housing supply (Sinai and Waldfogel, 2005; Eriksen and Rosenthal, 2010), and individual outcomes such as labour supply and income (Jacob and Ludwig, 2012). Importantly, these interventions may generate spillover effects on surrounding areas by altering neighborhood composition, amenities, and market segmentation, making their overall impact an empirical question of central policy relevance.

This project aims to study the local impact of social housing on a broad set of economic outcomes in the Netherlands, adopting a granular spatial perspective at the level of municipalities or neighborhoods. The analysis will examine how variation in the local intensity of social housing affects housing market dynamics (house prices, rents, and density), as well as individual outcomes such as employment, income, and labour market participation. In addition, the project is aimed at investigating whether social housing contributes to reducing inequality and poverty in targeted areas, and whether it generates spillover effects on nearby neighborhoods, capturing potential externalities across space.

The empirical analysis will rely on detailed administrative data from Statistics Netherlands (CBS), which provide rich information on housing, demographics, and socio-economic outcomes at a highly disaggregated geographical level, municipality- and neighborhood-level. The project will employ micro-econometric techniques to estimate both correlational and causal effects, including panel data methods and quasi-experimental approaches exploiting spatial and temporal variation in social housing provision. Furthermore, spatial econometric methods will be used to capture spillover effects and interactions across neighboring areas, allowing for a comprehensive assessment of both direct and indirect impacts of social housing policies.

Data source:

*Data on social housing and economic outcomes at the municipal level

https://opendata.cbs.nl/portal.html?_la=nl&_catalog=CBS&tableId=70072ned&_theme=244

*Data on social housing and economic outcomes at the neighborhood level

https://opendata.cbs.nl/portal.html?_la=nl&_catalog=CBS&tableId=84286NED&_theme=244

References:

*Eriksen, M. D., & Rosenthal, S. S. (2010). Crowd out effects of place-based subsidized rental housing: New evidence from the LIHTC program. *Journal of Public Economics*, 94(11-12), 953-966.

*Jacob, B. A., & Ludwig, J. (2012). The effects of housing assistance on labor supply: Evidence from a voucher lottery. *American Economic Review*, 102(1), 272-304.

*Koster, H. R., & Van Ommeren, J. (2019). Place-based policies and the housing market. *Review of Economics and Statistics*, 101(3), 400-414.

*Li, K., Su, S., Zhang, F., & Zhang, Y. (2025). Public housing and market segmentation: Evidence from the Joint Property Ownership scheme in China. *Regional Science and Urban Economics*, 104164.

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Factor pricing

Investment theory states that risk should be rewarded with higher expected returns. One of the first models to formalize this idea is the Capital Asset Pricing Model (CAPM). It states that expected returns are linear in a measure of risk:

$$E(R_{i,t}) = \lambda_0 + \beta'_i \lambda_f,$$

where $R_{i,t}$ is the return on asset i in period t , β_i is the regression coefficient from regressing the return on asset i on the k -dimensional factor f_t , λ_0 is the risk-free zero- β return, and λ_f is the k -dimensional vector of risk premia. In the CAPM, there is only one risk factor: the market return.

Using the Fama–MacBeth (1973) two-pass methodology, it was quickly discovered that one risk factor was too restrictive. Many other factors appeared to be priced, as reflected by their non-zero risk premia. See, for example, Fama and French (1993, 2015, 2018). This literature has expanded to the point where hundreds of potential risk factors have now been proposed, see Harvey et al. (2016) and Feng et al. (2020).

The proposed existence of so many factors that seem to explain asset returns raises the question of whether the estimation methodology is reliable. It turns out that the Fama–MacBeth two-pass procedure and the corresponding inference methods are quite sensitive and can lead to spurious conclusions, which may suggest that factors are priced when they are in fact not; see, for example, Kleibergen (2009) and Kleibergen and Zhan (2015, 2020). More recent work has therefore developed procedures that are more reliable.

Possible directions for the project:

1. A natural starting point is to become familiar with the Fama–MacBeth two-pass procedure and to implement it in practice. For this, students can use data from the Ken French data library: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html. The initial focus can be on US data, with the option to later consider other factors and/or other countries.
2. Replicate the fragility of the results for some well known papers, like, for example, Lettau and Ludvigson (2002) and more recent ones.
3. One can explore more recent, state-of-the-art inference procedures for analyzing risk premia, and compare them to the standard Fama–MacBeth approach.

References:

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Fair Machine Learning via projections

Supervisor: Floris Holstege

Machine learning (ML) algorithms now shape many parts of society, but their use can also create or reinforce inequality. A well-known example is the Dutch ‘toeslagenaffaire’ (Childcare Benefits Scandal), in which the tax office used an algorithm was more likely to flag people with a migrant background as misusing childcare benefits.¹ This case highlights the need for ML systems to satisfy fairness constraints.

Defining fairness. A common fairness notion in ML is ‘demographic parity’. This requires an algorithm’s predictions to be independent of a protected attribute. Let $\mathbf{x} \in \mathbb{R}^d$ denote the input, y the outcome of interest (for example, whether someone misused a subsidy), z a protected attribute (for example, race or gender), and $\hat{y}$ the algorithm’s prediction. Demographic parity requires $p(\hat{y} | z) = p(\hat{y})$. In other words, the probability of predicting misuse should not depend on race or gender.

Fairness via projection: One way to promote demographic parity is to transform the inputs $\mathbf{x}$ so that the transformed data has no covariance with the protected attribute z . We can do this with an orthogonal projection onto the null space of $\text{Cov}(\mathbf{x}, z) \in \mathbb{R}^d$. The transformed data is

$$\tilde{\mathbf{x}} = (\mathbf{I} - \mathbf{v}\mathbf{v}^T)\mathbf{x}, \text{ where } \mathbf{v} = \frac{\text{Cov}(\mathbf{x}, z)}{\|\text{Cov}(\mathbf{x}, z)\|}. \quad (1)$$

This guarantees $\text{Cov}(\tilde{\mathbf{x}}, z) = \mathbf{0}$. Previous work shows that this type of projection can help achieve demographic parity in linear regression and related linear models (see [Belrose et al. \(2023\)](#)).

In this project you will study how projection(s) can be used to achieve fairness constraints in ML. The starting point will be the projection introduced in equation 1. Some potential research questions are

- Under what conditions does the projection in equation 1 achieve demographic parity? and can we adapt it such that it achieves demographic parity under a wider range of conditions?
- The projection in equation 1 focuses on demographic parity. Can we augment it to achieve alternative fairness criteria, such as *equalized odds*?
- In practice, we often face a trade-off between fairness and other objectives, such as performance of our ML model. Can we formulate a projection that allows for this trade-off to be navigated?

Students are encouraged to come up with their own research question and ideas in addition to this.

Required knowledge: Students should have successfully completed *statistical learning* and have basic Python programming skills. An interest in linear algebra will also be helpful.

Resources:

- **Required before the first meeting:** read [Belrose et al. \(2023\)](#) (Sections 1–4), which introduces a projection that removes the covariance between $\mathbf{x}$ and z . Pay special attention to the difference with the projection in equation 1.
 - One key concept discussed in this paper is that of an *oblique projection*. For a short explanation of the difference between an orthogonal and oblique projection, see [this blog](#).
- For an illustration of the projection(s), students are encouraged to consult [this notebook](#).
- For an introduction to fair ML and different notions of fairness, see Chapter 3 of [Barocas et al. \(2023\)](#).

References

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¹A Dutch newspaper article about this example can be found [here](#). An English one [here](#).

Tabular Foundation Models in Econometrics

Supervisor: Floris Holstege

The widespread adoption of machine learning (ML) in society has been led by *foundation models*, models which have been *pre-trained* on enormous corpora of text and/or images (e.g. the GPT model series underlying ChatGPT). A standard ML approach is to take such a foundation model and fine-tune it on a dataset. In image and text classification, this approach has been shown to strongly outperform training a model from scratch on the respective dataset. Until recently, however, this strategy was not common for *tabular datasets* (for example, a table with firm-level data on their balance sheet). In recent years, several *tabular foundation models (TFMs)* have been introduced in order to bring the same foundation-model idea to tabular data. The goal of the project is to evaluate whether TFMs improve predictive performance relative to classical econometric methods, traditional ML models, and to understand the trade-offs involved.

Tabular foundation models (TFMs): You will work with large deep neural networks (DNNs) that have been pre-trained on tabular datasets, as well as synthetically generated data. The most popular TFM is TabPFN ([Hollmann et al., 2025](#)). You are encouraged to start with this TFM, and explore others (e.g. ([Qu et al., 2026](#))) if this you believe this to be necessary.

The application: You will compare a TFM to

- Classic econometric methods
- More ‘traditional’ ML methods

for a key prediction problem in econometrics that uses cross-sectional data. For example, you might predict the likelihood of credit default. You are encouraged to propose your own prediction problem, as long as it focuses on a relevant cross-sectional economic variable. Students are encouraged to think about how one can use TFMs in causal inference — for instance, in propensity score matching or in uncovering heterogeneous treatment effects.

The trade-offs: Beyond comparing predictive performance, you will evaluate additional dimensions that differentiate TFMs from classical econometric methods and traditional ML models. For example, you might analyze how interpretable each method is, or how robust it is to missing data.

Required knowledge: You should have successfully completed *statistical learning* and have strong Python programming skills.

Resources:

- **Required before the first meeting:** Read the paper introducing TabPFN ([Hollmann et al., 2025](#)). Focus on building intuition and understanding how the approach relates to standard econometric and machine learning methods, rather than on technical details.
- To learn how to implement a TFM, you should explore the Github repository for [TabPFN](#). These repositories provide step-by-step instructions for applying pretrained models to tabular prediction tasks.
- For a deeper understanding of how TFMs work, you can read [this blog](#). Do not worry if you do not understand every aspect of the model architecture.

References

- Hollmann, N., Müller, S., Purucker, L., Krishnakumar, A., Körfer, M., Hoo, S. B., Schirrmeister, R. T., and Hutter, F. (2025). Accurate predictions on small data with a tabular foundation model. *Nature*, 637(8045):319–326.
- Qu, J., Holzmüller, D., Varoquaux, G., and Morvan, M. L. (2026). Tabiclv2: A better, faster, scalable, and open tabular foundation model.

Improving the Forecasts of Agent-Based Models

Supervisor: Steven Hoekstra

Agent-based modeling has gained popularity in economic research because traditional equilibrium models struggle with non-linear events like financial crises, while newer agent-based models demonstrate comparable or superior forecasting accuracy. Software packages like the Bank of Italy's BeforeIT.jl have made this approach accessible to researchers and practitioners. See Axtell and Farmer (2025) and Dawid and Delli Gatti (2018) for an overview.

Project

You will use the BeforeIT.jl Julia package, based on Poledna et al. (2023), as your starting point. Your thesis involves proposing a focused improvement to the model and implementing it. Four directions are listed below, ordered by difficulty. You may also propose your own direction, subject to my approval. The evaluation method depends on the direction chosen: Directions 1 and 4 will be evaluated through a forecasting exercise (comparing RMSE of the base and improved model against real data); Direction 2 is a policy analysis (examining shock propagation under different specifications); Direction 3 is evaluated by how well the surrogate captures the ABM's behaviour. Data will be supplied.

Research Directions

Direction 1: Heterogeneous Expectations and Inflation Dynamics [Moderate]

Implement Brock and Hommes (1998) style heterogeneous expectations, where agents switch between forecasting rules based on past performance. This produces endogenous expectation densities and can generate inflation unanchoring dynamics. The modification is localised to the expectation formation module and the switching mechanism is well-documented. A natural application is to use the model to explain the 2022 inflation crisis: can heterogeneous expectations account for the sharp rise and subsequent decline in inflation better than the baseline model?

Direction 2: Energy Price Shocks and Network Propagation [Moderate]

Inject energy price shocks into specific sectors and track propagation through the 64-sector input-output network. In the current model, intermediate inputs are perfect substitutes; a key modelling choice is to replace this with a CES or Leontief production function and study how the degree of substitutability affects shock propagation. Compare the ABM's nonlinear amplification against linear Leontief multiplier predictions. This direction can be applied to the 2022 European energy crisis following Russia's invasion of Ukraine, or to the current energy price dynamics. The input-output structure is already in the model, and the Leontief benchmark provides a natural comparison.

Direction 3: Surrogate-Assisted Parameter Space Exploration [Challenging]

Train a machine learning surrogate to approximate the ABM following Lamperti et al. (2018), and test whether it remains accurate under policy interventions, as Dyer et al. (2023) showed observationally-trained surrogates can mispredict intervention effects. The evaluation criterion is how faithfully the surrogate reproduces the ABM's output across the parameter space. Since the main finding of Dyer et al. is that surrogates fail under interventions, your thesis should go beyond confirming this: quantify the failure modes (under which interventions, by how much) and propose or test a correction.

Direction 4: Simulation-Based Inference [Ambitious]

Apply neural posterior estimation to estimate posterior distributions over the model's parameters using the sbi Python package, following Dyer et al. (2024). This is the most technically demanding direction: it requires bridging Julia (ABM) and Python (sbi) and understanding neural density estimation. Only choose this if you are comfortable working across languages. A minimum viable thesis estimates posteriors for a subset of parameters; an excellent thesis validates the posteriors via simulation-based calibration and compares forecast performance against the baseline.

Prerequisites

This is a Julia project. All model work will be done in Julia. Python experience transfers well, but plan for a learning curve if you have not used Julia before.

Required before the first meeting:

1. Read Poledna et al. (2023) to understand the model architecture.
2. Work through lectures 1–5 of the Julia QuantEcon series (<https://julia.quantecon.org/intro.html>).
3. Install BeforeIT.jl and run at least one example from the repository (<https://github.com/bancaditalia/BeforeIT.jl>). Bring your output to the first meeting.

These are prerequisites, not recommendations. If you have not run the examples before our first meeting, we will spend the meeting on setup rather than research.

Recommended:

4. Work through lectures 6–10 of the Julia QuantEcon series.
5. For Direction 4: read the sbi package documentation.

References

- Axtell, R. L. and Farmer, J. D. (2025). Agent-Based Modeling in Economics and Finance: Past, Present, and Future. *Journal of Economic Literature*, 63(1):197–287.
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- Dyer, J., Cannon, P., Farmer, J. D., and Schmon, S. (2024). Black-box Bayesian inference for agent-based models. *Journal of Economic Dynamics and Control*, 161:104556.
- Dyer, J., Bishop, N., Felekis, Y., et al. (2023). Interventionally consistent surrogates for agent-based simulators. arXiv:2312.11158.
- Lamperti, F., Roventini, A., and Sani, A. (2018). Agent-based model calibration using machine learning surrogates. *Journal of Economic Dynamics and Control*, 90:366–389.
- Poledna, S., Miess, M. G., Hommes, C., and Rabitsch, K. (2023). Economic forecasting with an agent-based model. *European Economic Review*, 151:104306.

Livestock Production in Dry Rangelands Facing Climate Change

Supervisor: Marit Haringa

Introduction

African savannas and dry grasslands have been used as a natural resource for livestock production for thousands of years. Over the last couple of centuries, however, traditional livestock practices, characterised by nomadic, rotational herding across large distances, have been met with increased strain, by both human and climatic factors. Foreign settlement has divided traditionally common grazing lands into private, fenced off property, leaving a smaller slice of the pie for the traditional, nomadic pastoralists. Climate change has led to increased variability of rainfall, leading to droughts that may shift the rangeland into an irreversible, degraded state (e.g. a desert or a dry bushland). Human and climatic factors also interact: it has been observed that especially during droughts, intense conflicts may arise between traditional and commercial herdsmen.

In this project, you will model the management of a dry rangeland system. The system may contain traditional pastoralists, commercial farms, and everything in between. What are optimal levels of grazing intensity, slaughtering rate, and the optimal division of the land between agents, if such optima exist? How are these management levels affecting the probability of the system shifting to a degraded state? How are these results affected by increased variability of rainfall (or of other parameters)? Are some management strategies better suited for higher rainfall variability, and others better suited for low rainfall variability?

Example research questions

- Incorporating stochasticity of rainfall, fire intensity, or any other parameter in a pre-existing, deterministic model (e.g. in Haringa, Van Langevelde & Wagener, 2026). How resilient is optimal time-constant management against stochastic, instead of deterministic, shocks? Instead of optimal time-constant management, simpler rule-of-thumb management strategies can also be studied.
- Building a (simple) model representing the division of land and labor on dry rangelands. Traditional herdsmen are sometimes hired to help on commercial farms. Under what circumstances will commercial herdsmen drive out traditional nomadic practices?
- Modelling nomadic versus static livestock production, e.g. a nomadic system with two plots of grass and one herd, which is let to graze on one plot of grass at a time, versus a static system with only one plot of grass and one herd. Which situation is more resilient to e.g. droughts?

Methods, tools and skills

The aim of this project is to learn how to model real-life ecosystems using differential equations and how to simulate them using your own written computer programs. Affinity with programming (preferably Python, Matlab or Julia; I personally use Julia, and R is also allowed) is therefore an advantage, but not a necessity. The course Dynamical Systems is a good preparation for this project, but not a formal requirement. If you have not taken the Dynamical Systems course, the project will have a more general Mathematical Economics flavor, with perhaps some stochasticity/expectations mixed in. We won't use data, unless you provide it yourself.

References

Existing, deterministic, dry rangeland models with livestock management:

Haringa, M., Van Langevelde, F., & Wagener, F. O. O. (2026). Resilience of time-constant livestock management of savannas facing woody encroachment. [CeNDEF Working Papers, March 2026](#).

Langelaan, F., Van Langevelde, F., & Wagener, F. O. O. (2026). Resilience of a dry grazing system under different pastoral preferences. [CeNDEF Working Papers, February 2026](#).

Example of rangeland model with stochasticity (see e.g. eq. (11); also see general SDE literature):

Janssen, M. A., Anderies, J. M., & Walker, B. H. (2004). Robust strategies for managing rangelands with multiple stable attractors. *Journal of Environmental Economics and Management*, 47(1), 140-162.

General information about livestock practices in Africa:

Zantsi, S., Cloete, K., & Möhring, A. (2021). Productivity gap between commercial farmers and potential emerging farmers in South Africa: Implications for land redistribution policy. *Applied Animal Husbandry Rural Development*, 14, 22-31.

Greyling, J. C., Vink, N., & Mabaya, E. (2015). South Africa's agricultural sector twenty years after democracy (1994 to 2013). *Prof. Agric. Workers J.*, 3(1).

Human vs. Algorithm: Evaluating LLM-Based Title-Abstract Screening in a Clinical NLP Systematic Review

Systematic literature reviews underpin evidence-based medicine and health data science, but the screening phase is time-consuming, cognitively demanding, and susceptible to inter-rater variability. A typical clinical NLP review may require two independent screeners to assess several thousand title-abstract pairs, with disagreements resolved through discussion or a third reviewer. Automation of this process using large language models (LLMs) has attracted considerable recent interest, yet rigorous empirical evaluation of LLM screening accuracy against a human gold standard remains limited.

This proposal builds on ongoing systematic review of label-efficient methods for clinical concept extraction from electronic health records. The review applies a formally specified eligibility framework that defines clinical text scope, a four-criterion checklist for qualifying clinical concept extraction tasks (covering concept type, context resolution, output format, and evaluation alignment), and a precise definition of label-efficient versus non-label-efficient methods. This structured, multi-criterion framework forms a good starting point for evaluating whether LLMs can replicate human screening decisions.

Research Questions

Students may select and further develop one of the following research questions, subject to data availability and supervisor approval:

1. How accurately do LLMs replicate human inclusion/exclusion decisions in a clinical NLP systematic review when applied zero-shot with the full eligibility criteria provided in the prompt, as measured by sensitivity, specificity, and Cohen's kappa relative to the human gold standard?
2. Does multi-criterion screening accuracy differ systematically across models (e.g., GPT-4o, Claude Sonnet, Gemini 1.5 Pro), and can differences in performance be attributed to specific eligibility criteria on which models consistently disagree with human screeners?
3. To what extent does prompt design affect LLM screening accuracy, and is there a prompt specification that achieves sensitivity sufficient for use as a first-pass filter without unacceptable loss of eligible studies?

Methods and Data

The student will receive the candidate study set from the ongoing systematic review, comprising title-abstract pairs with human screening decisions (include, exclude, or uncertain) and, for excluded studies, the specific criterion on which exclusion was based. LLMs will be prompted to apply the eligibility framework and to return a structured decision (include/exclude) with a brief rationale. A minimum of three models will be evaluated: at least one from each of the major API providers (Anthropic, OpenAI, Google).

The student will work directly with data from the ongoing review under a data sharing agreement with the UNSW collaborating team.

Selected References

1. Syriani, E., David, I., & Kumar, G. (2023). Assessing the ability of ChatGPT to screen articles for systematic reviews. arXiv preprint arXiv:2307.06464.
2. Guo, E., Gupta, M., Deng, J., Park, Y. J., Paget, M., & Naugler, C. (2024). Automated paper screening for clinical reviews using large language models: data analysis study. *Journal of Medical Internet Research*, 26, e48996.
3. Agrawal, M. et al. (2022). Large language models are few-shot clinical information extractors. *Proceedings of EMNLP 2022*, 1998–2022.

Detecting Collusion Using Econometric and Machine Learning Methods: A Comparative Evaluation

Price-fixing and bid-rigging cartels inflict substantial welfare losses by elevating transaction prices above competitive levels over extended periods. Competition authorities have traditionally relied on documentary evidence, whistleblower testimony, and leniency applications to detect and prosecute collusion. Quantitative screening methods are increasingly used as a complement, enabling proactive enforcement without prior inside knowledge of the infringement.

Two broad families of detection methods have emerged in the literature. Econometric approaches apply structural break tests (e.g., Bai-Perron) and variance screens to price time series, exploiting the theoretical prediction that cartel coordination reduces price variance and produces level shifts at infringement onset and termination. Machine learning approaches, by contrast, treat detection as a supervised classification problem: algorithms are trained on labelled datasets containing confirmed collusive and competitive episodes and learn to identify statistical signatures of collusion across a range of distributional features. Wallimann et al. (2022) evaluate eleven ML classifiers across six procurement datasets. This proposal invites students to implement, compare, and critically evaluate both families of methods using publicly available data from documented cartel cases.

Research Questions

Students may select and further develop one of the following research questions, subject to data availability and supervisor approval:

1. Can Bai-Perron structural break tests reliably identify the start and end dates of documented price-fixing cartels in commodity markets, and how does detection accuracy vary with cartel duration and the number of participants?
2. Can Statistical Process Monitoring (SPM) methods help identify price-fixing cartels and how does detection accuracy vary with cartel duration and the number of participants?
3. To what extent do macroeconomic cost shocks (e.g., energy price movements) confound structural break detection in price-fixing cases, and can controlling for cost pass-through improve identification of the cartel period?

Methods and Data

Students will apply structural break tests (Bai-Perron, Chow, CUSUM) and/or supervised ML classifiers (logistic regression, random forest, gradient boosting) to price-level and price-variance series from publicly available commodity price databases and Eurostat producer price indices, using European Commission cartel decisions to establish infringement periods.

The framework of Wallimann et al. (2022), using the procurement datasets made available alongside that paper, can be the starting point, which covers confirmed collusive and competitive tender episodes from Brazil, Italy, Japan, Switzerland, and the United States. Model performance will be evaluated via cross-validated AUC, sensitivity, and specificity. Students should construct distributional features of bids (standard deviation, coefficient of variation, skewness, kurtosis, spread) as candidate predictors and assess feature importance. A direct comparison of ML and econometric method performance on a common holdout sample is encouraged where data permit.

Selected References

1. Wallimann, H., Imhof, D., & Huber, M. (2022). Collusion detection in public procurement auctions with machine learning algorithms. *Automation in Construction*, 133, 104047.
2. Bolotova, Y., Connor, J.M., & Miller, D.J. (2008). The impact of collusion on price behaviour: Empirical results from two recent cases. *International Journal of Industrial Organization*, 26(6), 1290–1307.
3. Bai, J., & Perron, P. (2003). Computation and analysis of multiple structural change models. *Journal of Applied Econometrics*, 18(1), 1–22.

Disclosure policies in case of a conflict of interests

(supervisor: Anita Kopányi-Peuker) – theory, simulation or data analysis

Professionals are often consulted to provide expert advice to uninformed decision makers (e.g., financial analysts about personal finances). In such cases, experts may have a conflict of interests in that they profit from their client taking an action that is not optimal for the client. Therefore, their advice may be biased against the interests of the decision maker.

One method to address this problem in the field is mandatory disclosure of such conflict. Previous theoretical (Li and Madarasz, 2008) and experimental (Cain et al., 2005) research have shown that in one-shot interactions, mandatory disclosure may harm the client because under disclosure the advisor biases their advice more than under nondisclosure, while the advisee does not correctly discount this. In a recent lab experiment, we compared different disclosure protocols in one-shot and repeated interactions between an advisor and a client (played by experimental participants). In particular, the experiment builds on the cheap-talk game (Crawford and Sobel, 1982), where the advisor knows the state of the world (integer up to 10), and can send a not payoff relevant message to the uninformed client. The client observes this message, and needs to take an action which determines both the advisor's and the client's payoff. The advisor can be of two types: they have the exact same payoffs as the client (and try to induce the action that matches the state of the world), or they want to induce a specific value for the client's action, irrespective of the state of the world (i.e., the advisor is biased).

The experiment implements three different disclosure protocols: no disclosure (the client does not know whether the advisor is biased), mandatory disclosure (the type of the advisor is known to the client) and voluntary disclosure (the advisor can decide to disclose their type to the client), as well as two relationship lengths: one-shot interaction (the same participants are not matched twice in a row), and repeated interactions (pairs are matched for longer).

Results show that disclosure helps reducing exaggeration by biased advisors both in frequency and magnitude and increasing trust in the message by clients. The data is available for students in this thesis project.

Possible research topics:

1. Game theoretical analysis of the corresponding one-shot / repeated game with social preferences or introducing lying / risk aversion in preferences.
2. Simulation analysis to find the most profitable strategy (e.g., by individual evolutionary learning, Arifovic et al., 2024; or horse-race between different simple strategies).
3. Estimation of behavioral rules from the experimental data.
4. Econometric analysis of the dynamics in the experimental decision making.

References:

- Arifovic, J., J. de Jong, A. Kopányi-Peuker (2024): Bank choice, bank runs, and coordination in the presence of two banks. *Journal of Economic Behavior & Organization*, Vol. 225, pp. 392-410. – in case the student is interested in point 2 (IEL)
- Cai, H. and Wang, J. T.-Y. (2006). Overcommunication in strategic information transmission games. *Games and Economic Behavior*, 56(1):7–36.
- Cain, D. M., G. Loewenstein, D. A. Moore (2005): The Dirt on Coming Clean: Perverse Effects of Disclosing Conflicts of Interest, *The Journal of Legal Studies*, Vol. 34, No. 1, pp. 1-25.
- Crawford, V. P., and J. Sobel (1982): Strategic Information Transmission, *Econometrica*, Vol. 50, No. 6, pp. 1431-1451.
- Li, M., and K. Madarasz (2008): When mandatory disclosure hurts: Expert advice and conflicting interests, *Journal of Economic Theory*, Vol. 139, 2008, pp. 47-74.

Import tariffs, market stability and welfare

(supervisor: Anita Kopányi-Peuker) – theory, simulation or data analysis

Although free trade is typically believed to maximize allocative efficiency, a decrease in barriers to trade might destabilize market dynamics and thereby be detrimental to welfare. In particular, recent theoretical work on coupled cobweb markets has shown that switching between markets by firms may lead to instability and perpetual price fluctuations that do not occur when these markets are isolated (see e.g. Dieci and Westerhoff, 2009, 2010; Tuinstra et al., 2014). Hence, there exists a trade-off between higher allocative efficiency at the steady state and higher price volatility. One implication of this is that import tariffs may enhance welfare by reducing volatility.

Previous laboratory experiments with isolated cobweb market showed quick convergence to the steady state with maximum allocative efficiency (e.g., Heemeijer et al., 2009; Bao et al., 2013). A recent experiment at the UvA extended those experiments by including two coupled markets (countries) with potential trade barriers in between in the form of import tariffs. There were two countries where the same homogeneous good is traded. Participants played the role of firms and their task was to decide how many units to supply in their home country as well as in the foreign country for 50 rounds. Production is described by one period lag, meaning that participants needed to decide about their supply without knowing the realized price yet. Their payoff depended on the profit their own firm made from trade. Treatments differed in the level of the import tariffs (0%, 20%, 30% and 80%). Results show that prices are more volatile with lower import tariffs. Furthermore, we find a non-monotonic effect of tariffs on welfare measured by consumer and producer surplus. The data from this experiment is available for this thesis project.

Possible research topics:

1. Theoretical analysis of a richer model with potentially more than two countries / welfare analysis.
2. Simulation analysis to find the most profitable strategy (e.g., by individual evolutionary learning, Arifovic et al., 2024).
3. Estimation of behavioral rules from the experimental data.
4. Inference of implied expectations of the participants in this experiment, and compare those to previous literature (e.g. Hommes et al., 2007, and Heemeijer et al., 2009).
5. Econometric analysis of the dynamics in the experimental decision making.

References:

- Arifovic, J., J. de Jong, A. Kopányi-Peuker (2024): Bank choice, bank runs, and coordination in the presence of two banks. *Journal of Economic Behavior & Organization*, Vol. 225, pp. 392-410. – in case the student is interested in point 2 (IEL)
- Bao, T., Duffy, J., and Hommes, C. (2013). Learning, forecasting and optimizing: An experimental study. *European Economic Review* 61: 186-204. (*)
- Dieci, R. and Westerhoff, F. (2009). Stability analysis of a cobweb model with market interactions. *Applied Mathematics and Computation* 215: 2011-2023.
- Dieci, R. and Westerhoff, F. (2010). Interacting cobweb markets. *Journal of Economic Behavior and Organization* 75: 461-481.
- Heemeijer, P., Hommes, C., Sonnemans, J., and Tuinstra, J. (2009). Price stability and volatility in markets with positive and negative expectations feedback: An experimental investigation. *Journal of Economic Dynamics and Control* 33:1052-1072.
- Hommes, C., Sonnemans, J., Tuinstra, J., and Van de Velden, H. (2007). Learning in cobweb experiments. *Macroeconomic Dynamics* 11 (S1): 8-33.
- Tuinstra, J., Wegener, M., and Westerhoff, F. (2014). Positive welfare effects of trade barriers in a dynamic partial equilibrium model. *Journal of Economic Dynamics and Control*, 48: 246-264. (*)

Bioeconomic Modelling and Tipping Points

Supervisor: Faan Langelaan

Introduction

In this project, we study the interaction between economic exploitation and the long-term sustainability of ecosystems. Many ecosystems provide resources that are valuable for human use, such as grazing land, fish stocks, or other biological resources. At the same time, these systems are governed by ecological processes and feedback mechanisms that determine whether they remain productive over time. Economic use can affect those dynamics and, in some cases, may push an ecosystem beyond a critical threshold.

Of special interest are ecosystems that feature *tipping points*. These are systems with critical thresholds at which a small change in environmental conditions, exploitation pressure, or species abundance can trigger a large and potentially irreversible shift to a different state or regime. Such shifts are often driven by positive feedback mechanisms, where an initial decline in ecosystem quality reinforces further decline. An example would be a grassland changing to a desert due to a drought, or a fish population disappearing due to overfishing. Tipping points occur in a wide range of ecological systems, including grasslands, coral reefs, fisheries, shallow lakes, forests and even some components of the climate system.

We measure the capacity of ecosystems to withstand such transitions by their *resilience*. In this project, we aim to study regime shifts and the ecosystem's resilience towards it under different ecological and economic model assumptions. There is wide range of possible systems that can be studied.

Example research questions

We are foremost interested in dynamical behavior, steady states and resilience of ecosystems under economic exploitation. Some different directions that could be explored are listed below, but many more subjects are possible, and you are encouraged to come up with your own questions.

- What is the effect of the inclusion of either seasonal periodicity or stochasticity in climatic conditions such as temperature and rainfall on the optimal management of ecosystems?
- Some papers have looked at the effect of including age-structuring of fish populations on the management of fisheries. Can we apply the same techniques to other systems: e.g. herds of livestock in grasslands?
- How do we calculate and evaluate the value of ecosystem resilience in an economic context?

Methods, tools and skills

The aim of this project is to learn more about the functioning of real-life ecosystems, how to represent them in mathematical models, and how to analyse these models using computer simulations. You will work with ordinary differential equations (ODEs) or stochastic differential equations (SDEs) to study ecosystem dynamics under different assumptions. Affinity with programming (preferably Python, Matlab or C++) is therefore an advantage, but not a necessity. Having followed the course **Dynamical Systems** will be an advantage in doing this project, but it is not a formal requirement.

Literature:

- Li, C. Z., Crépin, A.-S., & Folke, C. (2018). The Economics of Resilience. *International Review of Environmental and Resource Economics*, 11(4), 309–353.
- Vignal, T., Baudena, M., Mayor, A. G., & Sherratt, J. A. (2023). Impact of different destocking strategies on the resilience of dry rangelands. *Ecology and Evolution*, 13, e10102
- Weikard, H. P., & Hein, L. (2011). Efficient versus Sustainable Livestock Grazing in the Sahel. *Journal of Agricultural Economics*, 62(1), 153–171.
- Mumby, P. J., Hastings, A., & Edwards, H. J. (2007). Thresholds and the resilience of Caribbean coral reefs. *Nature*, 450, 98–101.

Assessing the Progress of EU Member States Towards a Circular Economy

Supervisor: Mentje Larney

The planned European Circular Economy Act, expected in 2026, is set to reinforce and harmonise EU rules on resource use, waste prevention, and product design, and is intended to accelerate the shift from a linear to a circular economy. This makes it a timely and relevant topic for a bachelor thesis.

The European Union's Circular Economy Action Plan (CEAP) - part of the European Green Deal - aims to “decouple economic growth from resource use” by promoting reuse, repair, recycling, and sustainable product design. To monitor progress, Eurostat and the European Environment Agency (EEA) provide indicators such as the circular material use rate (CMUR), material footprint, waste generation, and resource productivity. Recent empirical work shows that circular performance varies greatly across EU Member States and that improvements in circularity have been rather slow, raising questions about the effectiveness of EU-level circular economy policies.

This thesis will use publicly available EU data to assess whether progress on circular economy indicators is associated with reduced material use, and to what extent the CEAP has contributed to changes across Member States.

Possible research questions

The student will determine their own specific research question, but possible questions include:

1. To what extent is progress on circular economy indicators (e.g. CMUR, recycling rates, resource productivity) associated with reductions in material footprint per capita across EU Member States?
2. How have key circular economy indicators evolved across EU countries since 2010, and do “leader” and “laggard” countries differ systematically in their trajectories?
3. Is there evidence that the introduction of the CEAP has strengthened the decoupling between economic growth and material use in more circular economies?

Data

All data will be publicly available and primarily sourced from Eurostat and the EEA.

Suggested references as starting point

- Claudio-Quiroga, G., & Poza, C. (2024). Measuring the circular economy in Europe: Big differences among countries, great opportunities to converge. *Sustainable Development*, 32(5), 4707–4725.
- Remeikienė, R., Gasparėnienė, L. & Bankauskienė, J. (2025). Assessment of the circular economy indicators in the EU Member States. *Discov Sustain* 6, 611 (2025).
- Adriatik Kotorri, Llesh Lleshaj, Nertila Çika (2024). Empirical Analysis of Circular Economy in EU Countries for Environment Protection in Context of Resources Material and Waste Management. *Environment and Ecology Research*.

Determinantal Point Processes for Diverse Recommendations

Ravi Rajani

A common task in data science is to select products to recommend to a customer given their shopping basket. In many contexts – such as recommending videos to an active YouTube user [4] – one wants to show a diverse set of products. Originally identified as distributions arising in statistical physics, determinantal point processes (DPPs) are increasingly being used for this purpose [1, 2]. Roughly speaking, a DPP assigns a probability to a set of items proportional to the squared volume of the parallelepiped spanned by the items’ associated feature vectors.

In this project, you will select a DPP method for recommender systems described in the recent literature, reproduce one of the associated experiments in Python [3], and, either propose an extension to the method, or apply the method to a new dataset.

Prerequisites: Familiarity with machine learning, linear algebra, and Python programming.

References

- [1] Chen, L., Zhang, G., & Zhou, E. (2018). Fast greedy MAP inference for determinantal point process to improve recommendation diversity. In S. Bengio, H. M. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, & R. Garnett (Eds.), *Advances in Neural Information Processing Systems 31: Annual Conference on Neural Information Processing Systems 2018, NeurIPS 2018, December 3-8, 2018, Montréal, Canada* (pp. 5627–5638).
- [2] Gan, L., Nurbakova, D., Laporte, L., & Calabretto, S. (2020). Enhancing recommendation diversity using determinantal point processes on knowledge graphs. In J. X. Huang, Y. Chang, X. Cheng, J. Kamps, V. Murdock, J. Wen, & Y. Liu (Eds.), *Proceedings of the 43rd International ACM SIGIR conference on research and development in Information Retrieval, SIGIR 2020, Virtual Event, China, July 25-30, 2020* (pp. 2001–2004).: ACM.
- [3] Gautier, G., Polito, G., Bardenet, R., & Valko, M. (2019). DPPy: DPP Sampling with Python. *Journal of Machine Learning Research - Machine Learning Open Source Software (JMLR-MLOSS)*. Code at <http://github.com/guilgautier/DPPy/> Documentation at <http://dppy.readthedocs.io/>.
- [4] Wilhelm, M., Ramanathan, A., Bonomo, A., Jain, S., Chi, E. H., & Gillenwater, J. (2018). Practical diversified recommendations on YouTube with determinantal point processes. In A. Cuzzocrea, J. Allan, N. W. Paton, D. Srivastava, R. Agrawal, A. Z. Broder, M. J. Zaki, K. S. Candan, A. Labrinidis, A. Schuster, & H. Wang (Eds.), *Proceedings of the 27th ACM International Conference on Information and Knowledge Management, CIKM 2018, Torino, Italy, October 22-26, 2018* (pp. 2165–2173).: ACM.

Using option pricing theory to value investment opportunities

Daniël Roelfsema

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Subject:

Making sound investment decisions is a major challenge for companies. They may consider investing in a new factory, new machinery, or research and development (R&D). It is therefore crucial to estimate the added value of a potential investment and to assess the associated risks.

A widely used method for valuing investment projects is the Discounted Cash Flow (DCF) method. In this approach, the Net Present Value (NPV) of a project is computed to estimate the value of the investment opportunity, with project risk incorporated through the chosen discount rate. However, a key limitation of this methodology is its deterministic nature: the uncertainty and stochastic behaviour of future cash flows are not fully captured. In addition, the traditional DCF framework has difficulty reflecting the staged nature of many investments, such as R&D projects, which may progress to a highly successful “next stage” or fail altogether, resulting in a total loss of the invested capital.

In their article [1], Song, Xie, Ching and Siu propose an alternative approach that is gaining popularity: using real option valuation theory to assess investment opportunities in high-tech companies. The real options approach adapts techniques used to value financial put and call options to determine the value of an investment opportunity. This typically involves applying the well-known Black–Scholes option pricing formula or, perhaps more appropriately, the binomial tree model for pricing American-style options

In this project, you will develop a mathematical model to value an investment opportunity of your choice using real option pricing theory and conduct a risk analysis of this project. To evaluate the financial attractiveness of the opportunity, you will use (Monte Carlo) simulation techniques.

To be successful in this research project, some background and interest in both finance and probability theory are essential. You will take your own route in developing the model, while drawing on the existing literature to justify your modelling choices

References:

- [1] Song, N., Xie, Y., Ching, W. K., & Siu, T. K. (2017). A real option approach for investment opportunity valuation. *J. Ind. Manag. Optim.*, 13(3), 1213-1235.
(<https://pdfs.semanticscholar.org/2601/c7e1bc0a559ee3b2fa1251591c673a5e655a.pdf>)
- [2] Karnavas, C. (2024). Stochastic Analysis in Investment Project Appraisal. *International Journal of Engineering Inventions*, 13(3), pages 16-22.
(https://www.researchgate.net/publication/398110275_Stochastic_Analysis_in_Investment_Project_Appraisal)
- [3] Luehrman, T. A. (1998). Investment opportunities as real options: Getting started on the numbers. *Harvard Business Review*, July-August 1998
(<https://people.stern.nyu.edu/adamodar/pdfiles/articles/InvestmentsasOptions.pdf>)

(Time) Varying Parameters in Econometrics

Supervisor: Mario Rothfelder

Background: With economic systems being ever changing, estimating econometric models with (time) constant parameters has been questioned from the early days of the field, cf. the debate between John M. Keynes and Jan Tinbergen. Arguably, several reasons such as policy changes, exogenous shifts, economic/technological development and others may, at the very least, alter the parameters of an economic model. Consequently, advice based on “outdated” models may prove itself harmful rather than helpful or may be misguided in real world scenarios. Hence, accounting for changing model parameters is crucial to allow for the model’s reliability and hence, a plethora of models/approaches for (time) varying parameters in different fields and circumstances have been proposed in the past: Structural break models, Threshold Models, Random coefficient models and GAS-models, to name a few.

Objective: Within this project you will gain a deeper understanding of a (time) varying parameter model of your choice. Using simulated or real data, potential topics are

- Using a *threshold model*, investigate the effects of government spending on GDP growth via local projections
- Investigate potential changes in identification strength in linear IV models via *structural break models*
- Investigate instabilities at the boundary of the search grid in *threshold* or *structural break models*
- Investigate what the effects of recent world political events on several important commodities are
- Investigate whether monetary policy reaction functions are different during the tenure of different chairmen
- Investigate what the consequences of model misspecifications are

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Measuring Associations and Evaluating Forecasts of Categorical Variables

Supervisor: Andrei Sirchenko (personal webpage: www.sirchenko.info)

Capacity: 1 student

Do you like coding and developing computational software? Would you like to submit a paper to an international journal? You can do both if you choose this project!

The goal of this project is to develop a new command for Stata and submit the results as a research article to *The Stata Journal*. The command will be implemented using Mata, Stata's matrix programming language, which is similar in style to C, Java, and R.

This is my third call for collaboration with bachelor students on a Stata Journal paper. Two previous projects have already resulted in published articles (Dale and Sirchenko, 2021; Huismans, Nijenhuis, and Sirchenko, 2022). Importantly, these students had no prior experience with Mata before starting their bachelor theses! At this stage, a beta version of the command already exists and needs to be completed, tested, and documented. We will present the paper at the German Stata Conference in Munich on June 19.

Categorical variables (qualitative or discrete-valued numerical variables) are widely used across many fields, including economics, statistics, computer science, machine learning, and the social and life sciences. Measuring the strength of association between categorical variables and evaluating the accuracy of forecasts of categorical outcomes are important methodological problems.

Over the past two centuries, researchers have proposed many scalar statistics to quantify association, agreement, dependence, or predictive accuracy between categorical variables. Examples include measures of agreement, similarity, correlation, dependence, predictability, and reliability. For discussions of their properties and relationships, see, for example, Choi et al. (2010), Gneiting and Raftery (2007), and Goodman (2000).

A new Stata command will compute a comprehensive collection of over 200 quantitative measures of association and forecast accuracy for categorical variables (dichotomous, and nominal and ordinal polytomous). In addition to overall association measures, the command will compute category-specific diagnostic metrics for each observed category, as well as their macro and weighted averages.

For further information please email me: andrei.sirchenko@gmail.com.

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Do Discreteness and Regime Switching Matter for Monetary Policy?

Modeling Policy Interest Rate Decisions

Supervisor: Andrei Sirchenko (personal webpage: www.sirchenko.info)

Capacity: 1-10 students

Monetary policy rules attract substantial interest among economists, policymakers, and financial markets. Market participants closely monitor central banks to predict future policy decisions, while academic research frequently models monetary policy using policy reaction functions. Central banks themselves often consider various interest rate rules when setting policy.

A large body of empirical literature models monetary policy decisions using Taylor-type policy rules, which relate the central bank policy rate to macroeconomic variables such as inflation and output gaps. These rules are typically estimated using linear regression models with continuous dependent variables. However, in practice central banks adjust their policy interest rates in discrete steps, most commonly in increments of 25 basis points. This discreteness has been a standard feature of monetary policy in many countries for at least the past three decades.

Because of this feature, treating policy rates as continuous variables may be inappropriate. The discrete and censored nature of policy rate changes suggests the use of econometric methods designed for ordinal outcomes, such as ordered discrete-choice models (see Hamilton and Jordà, 2002; Monokroussos, 2011). In addition, monetary policy decisions often differ across phases of the business cycle. This motivates the use of regime-switching models, which allow model parameters and dynamics to vary across different economic regimes (Sirchenko, 2020; Dale and Sirchenko, 2021; Seibert et al., 2021; Huismans et al., 2022).

The goal of this project is to investigate whether accounting for discreteness and regime switching improves empirical identification of monetary policy behavior.

Discrete-choice models estimate probabilities of specific policy actions, while linear OLS regression models estimate the expected value of a continuous dependent variable. As a result, traditional measures of model fit used for OLS regressions cannot be directly applied to discrete-choice models, and vice versa. It seems also impossible to construct a formal statistical test (e.g., based on the likelihood) to compare them.

Are there any appropriate ways to compare them? You will explore some of them modelling central bank policy decisions in one of the countries such as the USA, UK, EU, Poland, Australia, Sweden, Japan, etc.

References

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Growth-at-Risk

Supervisor: Ekaterina Ugulava

Recent developments in macroeconomic analysis emphasise a risk-management approach to policy-making, whereby policymakers are concerned not only with expected economic outcomes but also with the distribution of risks surrounding them, in particular the likelihood of severe downturns. This has led to the emergence of the Growth-at-Risk (GaR) framework, which models the entire conditional distribution of future GDP growth, rather than focusing only on its mean. This framework originates from (Adrian, Boyarchenko, & Giannone, 2019), who establish the empirical usefulness of quantile regression for modelling and particularly forecasting the conditional density of US GDP growth. Their findings show that deteriorating financial conditions, as captured by the Chicago Fed’s National Financial Conditions Index (NFCI), have an asymmetric effect on GDP growth. In particular, weak financial conditions significantly increase downside risks to growth.

The following set of questions could be investigated in relation to this topic (other questions are also possible):

- Instead of relying on existing indices (e.g., NFCI), one may construct a financial conditions factor using a Dynamic Factor Model based on a set of financial indicators. This factor can then be used as a predictor in a quantile regression model for future GDP growth. Does this factor provide additional predictive power relative to an existing index?
- The GaR framework firstly estimates several quantiles via quantile regression model and then fits a parametric distribution (e.g., Skewed-t) to obtain a full predictive density. The downside is that this approach imposes a strong distributional assumption on density. An alternative is to construct the predictive distribution using nonparametric methods through interpolation and extrapolation techniques. Does imposing a parametric distribution affect the assessment of downside risks?
- An alternative approach to quantile regression is to model macroeconomic risks using volatility models, such as GARCH. These models generate a conditional distribution by using an assumption on error distribution and estimated conditional volatility. Do volatility models adequately capture tail risks? Do quantile-based and GARCH-based approaches produce systematically different estimates of downside risks?

References

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Link your network with SAR models

supervisor: *Kees Jan van Garderen*

LinkedIn and Facebook are important and/or popular networks that play important roles in business, the public sector, and private lives. When economic agents, like firms and consumers, interact, they form a network and make decisions depending on the business- or social network in which they operate. They will observe and take into account what other agents do in this network. The usual assumption of independent observations is unrealistic in these cases and will be violated. Similarly with house prices, which are directly affected by prices in a neighbourhood. Spatial autoregressive models model these relations explicitly through correlations. There are two simple popular models.

Let $W : (N \times N)$ be the so called spatial correlation matrix, assumed fixed and known, such that

$$\begin{aligned} \text{Model 1} & : y = X\beta + \varepsilon, \\ & : \varepsilon = \lambda W\varepsilon + \sigma u, \\ \text{Model 2} & : y = \lambda Wy + X\beta + \sigma u, \end{aligned}$$

with $u \sim \mathcal{N}(0, I_N)$ in both cases.

Model 1 models spatial correlation in the disturbances, whereas *Model 2* models the spatial dependence of the endogenous variables. This is similar to modeling time series via the disturbances, as in: $y_t = x_t'\beta + u_t$ with $u_t = \lambda u_{t-1} + \varepsilon_t$, or via a dynamic specification: $y_t = \lambda y_{t-1} + x_t'\beta + u_t$.

Model 2 without any regressors X is called a *pure* Spatial Autoregressive (SAR) model. The matrix W is generally assumed known in the literature and there are many alternatives and the appropriateness will depend on the network that is being analysed.

The parameter of interest in this model is λ . If $\lambda = 0$ then there is no spatial correlation. This provides an interesting null hypothesis to test. Second, it is of interest to provide confidence intervals for λ , precisely because we think λ is not 0.

There are various interesting applications and theoretical aspects of these models. This research proposal invites you to formulate and investigate a research question, including for instance:

1. empirical questions on how important this is for data that you find interesting,
2. the effects of sample size,
3. the impact of uncertainty σ ,
4. type of spatial correlation, i.e. the choice of W or its parameter ρ ,
5. compare tests for the presence of spatial autoregression,
6. different types of confidence intervals.

Software. Mathematica, R or any other programme that is convenient

References

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Inequality Measurement and Inference

Supervisor: *Kees Jan van Garderen*

Economic inequality occurs in many different settings. The Deaton Review at the IFS in London has recently considered wealth-, geographical-, gender-, educational-, benefit-, racial-, and various other inequalities. In order to answer questions whether inequality increased in the Netherlands in the last 25 years, or if inequality is larger in the US than in Canada, etc. we need a measure of inequality for comparing different situations. We also need an idea of uncertainty in these measures if we want to conclude that differences are significant. There are many different measures of inequality. They measure the distribution of income or other inputs and resources. Well known examples are the Gini coefficient, the Atkinson index, the (two) Theil indices, and the (Hirschman-) Herfindahl index which plays an important role in industrial organization. The last three are special cases of a family of inequality indices called the Generalized Entropy indices, and the Atkinson index is ordinally related. The class of Generalized Entropy indices is defined for any real α by

$$I(\alpha; F) = \begin{cases} \frac{1}{\alpha^2 - \alpha} \left[\frac{\mu_\alpha(F)}{\mu_1(F)^\alpha} - 1 \right] & \text{for } \alpha \neq \{0, 1\} \\ - \int \log\left(\frac{x}{\mu_1(F)}\right) dF(x) & \text{for } \alpha = 0 \\ \int \frac{x}{\mu_1(F)} \log\left(\frac{x}{\mu_1(F)}\right) dF(x) & \text{for } \alpha = 1 \end{cases}$$

where α is a sensitivity parameter, F is the income distribution, and $\mu_\alpha(F) = \int x^\alpha dF(x)$ is the moment functional, and we will assume incomes to be positive. The index is continuous in α . The larger the parameter α , the larger is the sensitivity of the inequality index to the upper tail of the income distribution. For an extensive discussion of the properties of the GE index see Cowell (1977, 1980, 2000).

In practice one makes a choice for α and calculates the GE index for different populations. These measures are then compared, but they are based on samples and have varying degrees of uncertainty and associated confidence intervals. Confidence intervals are generally based on simple first order approximations using the delta method. These confidence intervals can be very unreliable with coverage rates of less than 85% for 95% nominal confidence intervals. See Schluter and Van Garderen (2009) who propose a higher order correction to obtain more reliable confidence intervals, but is rather involved. We could use simulation methods instead.

An alternative is to use maximum likelihood procedures to construct confidence intervals. They provide very reliable confidence intervals as long as the right income distribution is assumed. If the wrong distribution is assumed the confidence intervals can go dramatically wrong. In practice we do not know what the true income distribution is and one runs the risk of making very large mistakes.

This project allows for two broadly different research question

- Empirical: use relevant data to answer an interesting inequality question
- Methodological: investigate which method of estimation and inference is more reliable.

Software. Mathematica, R or any other programme that is convenient

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Using Machine Learning Methods to Estimate the Gender Wage - Supervisor Noud van Giersbergen

For decades, empirical economics relied on methods such as ordinary least squares (OLS) and instrumental variables to estimate structural relationships. These tools remain powerful in low-dimensional settings, but they become less reliable when researchers face very large datasets, many covariates, complex interactions, and non-linear relationships. In such contexts, standard econometric models are vulnerable to functional form misspecification and omitted variable bias.

Machine learning, by contrast, is designed to handle high-dimensional and non-linear data structures. Algorithms such as random forests, gradient boosting, lasso, and neural networks can detect patterns and optimize predictive performance. However, off-the-shelf ML is not automatically suitable for causal inference. Its reliance on regularization and shrinkage introduces bias in the parameter of interest, which makes standard inference invalid when the goal is to estimate a causal effect rather than merely predict an outcome.

The thesis project therefore centers on the field of **Causal Machine Learning**, which combines the flexibility of ML with the identification logic of econometrics. Its central methodological anchor is **Double/Debiased Machine Learning (DML)**, as formalized by Chernozhukov et al. (2018). DML allows researchers to use flexible ML tools to estimate nuisance functions while preserving valid statistical inference for a low-dimensional causal parameter.

Core Empirical Application

The shared empirical setting for the proposed thesis cohort is the estimation of the **gender wage gap** using observational microdata from the **American Community Survey (ACS)**¹. The gender wage gap is traditionally studied using Mincer-style wage regressions with controls for education, experience, occupation, and other characteristics. Yet these models often impose simplistic linear assumptions and cannot fully account for high-dimensional heterogeneity.

The proposal treats gender as the treatment variable and log hourly wages as the outcome. The causal parameter of interest is the effect of being female on wages, conditional on a large vector of covariates such as education, age, geography, occupation, and demographic background. The underlying partially linear model is:

$$\begin{aligned} Y_i &= \theta_0 D_i + g_0(X_i) + U_i, \\ D_i &= m_0(X_i) + V_i, \end{aligned}$$

where Y_i is the outcome (log wages), D_i is the treatment indicator (female), X_i is the high-dimensional control vector, θ_0 is the target causal effect, $g_0(X_i)$ and $m_0(X_i)$ are nuisance functions. The control vector may include variables related to human capital and demographics (e.g., educational attainment, age and experience, racial identity, and other individual characteristics), job and labor market characteristics (e.g., occupational classifications, industry identifiers, and work arrangements), and geographic factors (e.g., location).

A naive ML regression of wages on gender and controls would typically shrink the treatment coefficient and generate biased estimates. DML corrects this through two key mechanisms. First, **Neyman orthogonality** constructs a score that is locally insensitive to small errors in the estimation of nuisance functions. Second, **sample splitting and cross-fitting** prevent overfitting by ensuring that the same observations are not used both to train nuisance models and estimate the final causal parameter. Together, these steps restore valid $\sqrt{N}$ -consistent inference.

¹<https://usa.ipums.org/usa/>

Four Thesis Twists

Twist 1: Algorithmic Sensitivity and Ensemble Learning. How sensitive are the DML estimates to the choice of ML algorithm used to estimate nuisance functions? You compare methods such as lasso, ridge, random forests, boosted trees, and neural networks, and then implements *stacking* or *super learner* ensembles. The expected result is that while point estimates of the causal wage gap should remain relatively stable due to orthogonality, the precision of estimates may vary with the quality of nuisance prediction.

Twist 2: Heterogeneous Treatment Effects and Causal Forests. This project moves beyond the average treatment effect and studies heterogeneity in the gender wage gap across groups. Using causal forests, you estimate group average treatment effects and possibly individual treatment effects. This is especially relevant for intersectionality, since the wage penalty may vary across race, age, education, or region. The expected contribution is a richer map of how wage inequality is distributed across the population.

Twist 3: Causal Mediation and Occupational Sorting. What are the mechanisms behind the wage gap by separating the *direct effect* of gender on wages from the *indirect effect* operating through occupational sorting? Occupation is not just a control variable; it is itself shaped by gendered social processes. You therefore applies causal mediation analysis within a DML framework to estimate the extent to which the wage gap reflects within-job discrimination versus structural sorting into lower-paid occupations.

Twist 4: High-Cardinality Features and NLP. This twist focuses on feature engineering. Traditional econometric models handle large numbers of categories poorly, especially when occupation, industry, or location variables are highly granular. You therefore applies modern ML tools such as target encoding, embeddings, or TF-IDF to transform high-cardinality categorical or textual information into dense numerical features. These richer representations are then used in the DML pipeline to improve adjustment for observed heterogeneity.

References

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High-Dimensional Covariance Matrix Estimation and Portfolio Construction - Supervisor Noud van Giersbergen

The foundation of modern quantitative finance and asset allocation rests on the mean–variance portfolio optimization framework introduced by Harry Markowitz in 1952. This theory states that an investor can construct an optimal portfolio by minimizing variance for a given level of expected return or, equivalently, by maximizing expected return for a given level of risk. In the absence of specific constraints or reliable forecasts of expected returns, this framework yields the **Global Minimum Variance (GMV)** portfolio, which relies exclusively on the covariance matrix of asset returns to determine optimal asset weights.

However, the practical implementation of mean–variance optimization has historically been plagued by severe statistical and econometric limitations. The analytical solution for the optimal portfolio weights requires the inversion of the covariance matrix, often referred to as the **precision matrix**. In a real-world setting, the true population covariance matrix is unobservable and must be estimated from finite historical time-series data. This standard methodology breaks down completely in high-dimensional settings, specifically when the number of assets (N) is large and particularly when it exceeds the number of historical observations (T). In the $N > T$ regime, the sample covariance matrix becomes mathematically singular. Because its rank is at most $T - 1$, it is not of full rank; hence, its inverse does not exist, rendering the traditional Markowitz solution infeasible. Even in scenarios where N is slightly less than T (e.g., $N/T \rightarrow c \in (0, 1)$), the sample covariance matrix is notoriously ill-conditioned. The inversion of an ill-conditioned matrix acts as an “error maximizer,” aggressively amplifying the estimation noise present in the sample.

The quadratic optimizer inherently exploits these errors by assigning excessively large positive weights to assets with spuriously low estimated variances or negative covariances, and excessively large negative weights to assets with spuriously high estimated variances. Consequently, sample-based mean–variance portfolios perform poorly out of sample. This phenomenon is so severe that the influential study by DeMiguel, Garlappi, and Uppal (2009) demonstrates that a naive $1/N$ equally weighted portfolio—which completely ignores covariance information—often outperforms optimally diversified Markowitz portfolios in out-of-sample tests.

This project is based on the influential econometric paper by Ledoit and Wolf (2017), titled *Nonlinear Shrinkage of the Covariance Matrix for Portfolio Selection: Markowitz Meets Goldilocks*.

Let r_t be an N -dimensional vector of stationary asset returns at time t , with expected return vector μ and covariance matrix Σ . The objective is to find the portfolio weight vector w that minimizes the portfolio variance ($w^T \Sigma w$), subject to the full-investment constraint:

$$\min_w w^T \Sigma w \quad \text{subject to} \quad w^T \mathbf{1}_N = 1 \quad (1)$$

The analytical solution for the GMV portfolio is:

$$w_{GMV} = \frac{\Sigma^{-1} \mathbf{1}_N}{\mathbf{1}_N^T \Sigma^{-1} \mathbf{1}_N} \quad (2)$$

When $N > T$, the traditional sample covariance estimator $S = \frac{1}{T-1} \sum_{t=1}^T (r_t - \bar{r})(r_t - \bar{r})^T$ is singular. Ledoit and Wolf (2017) address this issue by applying a nonlinear transformation to the eigenvalues of S . Let the spectral decomposition be $S = U \Lambda U^T$. The nonlinear shrinkage estimator takes the form:

$$\hat{\Sigma}_{NL} = U \tilde{\Lambda} U^T \quad (3)$$

where $\tilde{\Lambda}$ is a diagonal matrix of adjusted eigenvalues.

Freely Available Data Sources

Table 1: Data sources

Data Component	Description and Scope	Recommended Source
Asset Universe	Daily adjusted prices ($N \approx 500$)	Yahoo Finance (yfinance/quantmod)
Index Constituents	Historical S&P 500 lists	Wikipedia / Mendeley Data
Risk-Free Rate	One-month U.S. Treasury bill rate	Kenneth R. French Data Library ^a
Structural Factors	Fama–French 3 or 5-factor returns	Kenneth R. French Data Library ^b

^a Available at: Kenneth R. French Data Library.

^b Factor return series can be downloaded from the same source.

Five Thesis Twists

Twist 1: Linear vs. Non-Linear Shrinkage. Compares the 2004 linear shrinkage $\hat{\Sigma}_{Linear} = \alpha F + (1 - \alpha)S$ against the 2017 adaptive non-linear spectrum adjustment.

Twist 2: Factor-Based Estimation (POET). Decomposes the matrix into systemic and idiosyncratic components: $\Sigma = B\Sigma_f B^T + \Sigma_u$, applying thresholding to the latter.

Twist 3: Precision Matrix Sparsity (Glasso). Directly estimates $\Omega = \Sigma^{-1}$ using L_1 penalization:

$$\hat{\Omega}_{Glasso} = \arg \max_{\Omega \succ 0} (\log \det \Omega - \text{tr}(S\Omega) - \lambda \|\Omega\|_1) \quad (4)$$

Twist 4: Random Matrix Theory (RMT). Uses the Marchenko-Pastur distribution to separate signal from noise, clipping eigenvalues below the theoretical bound $\lambda_+ = \sigma^2(1 + \sqrt{N/T})^2$.

Twist 5: Hierarchical Risk Parity (HRP). A machine-learning approach that uses tree clustering and recursive bisection to bypass matrix inversion entirely.

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Comparing estimation techniques for correction sample selectivity.

Supervisor: Hans van Ophem.

#Students: 5-10

Research question:

In Garlick and Hyman (2022) several semiparametric estimators of a selectivity model are estimated. They chose for a quasi-experimental approach that has the disadvantage that biases can be studied only under the assumption that the underlying process remains stable in time. In a pure experimental approach this assumption can be avoided and it is interesting to investigate whether the conclusions found in Garlick and Hyman (2022) depend on this strong assumption. Alternatively, the main conclusion that OLS (i.e. ignoring selectivity) works just as well as more complex semiparametric estimation methods (e.g. the method of Newey, 1999). What Garlick and Hyman (2022) also find is that parametric estimation methods are the worst methods to use (variants of the method of Heckman (1979)). Hussinger (2008) comes more or less to the same conclusion (although given her interest she does not want to state this conclusion but it is clear from the estimation results provided). These conclusions can also follow from the specific problem these authors investigated or the specific data used. Is the conclusion that selectivity correction methods work no better than OLS (without correction) also true in a different empirical setting? This research can be carried out on real data and simulated data and different combinations of estimation techniques can be employed.

Data:

- Real data from e.g. Centerdata, NLSY, data from Fabrizio and Tsohman (2014) etc. or
- Simulated data.

Issues:

Selectivity, bias, alternative estimation methods, semiparametrics.

Literature:

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Sample selection correction in a Poisson regression model

Supervisor: Hans van Ophem.

#Students: 5-10

Research question:

Bergeot and Tenand (2023) investigate the effect of informal care on care costs, using a control function approach that consists of a two-stage estimation procedure. The dependent variable is a count that is estimated with a Poisson model. One of the dummy explanatory variables is endogenous, and the parametric estimation method proposed in Wooldridge (2015) is applied to estimate the model consistently. However, the method is parametric (for sure in the first step and debatable in the second step) and assumes a normal distributed error term (first step probit estimation). Applying the semiparametric methods of Cosslett or Newey (discussed in Hussinger (2008) and Garlick and Hyman (2022)) might be applicable to the Bergeot and Tenand setting. To investigate this a Monte Carlo analysis can be employed or an analysis on real data could be carried out to compare different estimation methods.

Data:

- Real data from e.g. Centerdata, NLSY, data from Fabrizio and Tsoolman (2014) etc. or
- Simulated data.

Issues:

Poisson model, Quasi maximum likelihood, selectivity, bias, alternative estimation methods, semi-parametrics.

Literature:

- *Bergeot, J., and M. Tenand, Does informal care delay nursing home entry? Evidence from Dutch linked survey and administrative data, *Journal of Health Economics*, 2023 (vol. 92), pp. 1-17.
- Fabrizio, K.R., and U. Tsoolmon, An empirical examination of the procyclicality of R&D investment and innovation, *The Review of Economics and Statistics*, 2014, pp. 662-675.
- *Garlick, R., and J. Hyman, Quasi-experimental evaluation of alternative sample selection corrections, *Journal of Business & Economic Statistics*, 2022, pp. 950-964.
- Hussinger, K., R&D and subsidies at the firm level: An application of parametric and semi-parametric two-step selection models, *Journal of Applied Econometrics*, 2008, pp. 729-747.
- Wooldridge, J.M., Quasi-maximum likelihood estimation and testing for nonlinear models with endogenous explanatory variables. *Journal of Econometrics* 2014 (vol 182-1), pp. 226-234.
- Wooldridge, J.M., Control function methods in applied econometrics. *Journal of Human Resources*, 2015 (vol. 50-2), pp. 420-445.

How Robust Is Double Machine Learning?

Supervisor: Nadja van 't Hoff

In many empirical studies, researchers want to estimate the effect of a treatment or policy while controlling for many other variables. Recent econometric approaches combine machine learning with methods for causal analysis to handle such settings. A prominent example is Double Machine Learning (DML), which uses machine learning algorithms to model the relationship between variables, treatment, and outcomes while still allowing reliable statistical inference.

DML allows researchers to use different machine learning methods, such as Lasso, ridge regression, random forests, or boosting. However, these algorithms rely on different assumptions about the data. As a result, estimated effects may be sensitive to the choice of method and to characteristics of the data.

You will estimate treatment effects using DML with different machine learning methods and compare the resulting estimates. The robustness of these estimates will be studied either through simulation experiments, where key features of the data are systematically varied, or through an empirical analysis comparing treatment effects across methods in an applied dataset.

Possible research questions:

- How sensitive are DML estimates to the choice of machine learning method under different data conditions, such as sparse vs. dense relationships, nonlinearities, high-dimensional settings, tuning choices, and computational constraints?
- How large is the bias when sparsity-based learners omit variables with small but relevant effects? How does correlation among variables influence the bias? How does this depend on the number and strength of such variables? Can non-sparse learners mitigate the problem?
- How much do estimated treatment effects vary across different machine learning methods or tuning parameters in an applied dataset? Do these differences affect policy recommendations?

Literature on DML:

- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1), C1–C68.
- Chernozhukov, V., Hansen, C., Kallus, N., Spindler, M., & Syrgkanis, V. (2024). Applied causal inference powered by ML and AI. *arXiv preprint arXiv:2403.02467*. (mainly Chapter 9)

Examples of Empirical Studies and Datasets:

- **Job training and earnings:** LaLonde, R. J. (1986). Evaluating the econometric evaluations of training programs with experimental data. *American Economic Review*, 76(4), 604–620.
- **Class size and student achievement:** Krueger, A. B. (1999). *Experimental estimates of education production functions*. *Quarterly Journal of Economics*.
- **Health behaviors and health outcomes:** Pruijm, R. (2015). *NHANES: Data from the US National Health and Nutrition Examination Study* (R package version 2.1.0).

Heterogeneous Peer Effects under Social Network Interactions

Supervisor: Lina Zhang

Many studies of causal effects assume that individuals do not interact with one another. However, this is a strong assumption that may not hold in real-world settings. For example, when studying the effectiveness of vaccines, an individual's outcome may depend not only on their own vaccination status but also on the vaccination status of others with whom they interact daily. Such interactions can generate peer effects, where the behavior or treatment of others influences an individual's outcome. Understanding whether and how peers affect outcomes is therefore of great interest to many researchers.

Many existing methods for investigating peer effects rely on simple linear models. While linear models are easy to estimate and interpret, their simplicity comes at the cost of limited flexibility. In particular, linear models may suffer from model misspecification and may fail to capture the potential heterogeneity of peer effects. For instance, an individual with no vaccinated peers may benefit more from being vaccinated than someone who already has many vaccinated peers. A linear model may not adequately capture such heterogeneous effects.

In this thesis, students will analyze the heterogeneity of peer effects in social experiments by adopting the method proposed by Leung (2020). This method uses a nonparametric estimation approach that has been shown to possess desirable asymptotic properties. Students will apply the method of Leung (2020) to one of the empirical studies listed below, and compare the resulting nonparametric estimates with those obtained using the linear models employed in the original papers. The following empirical studies are available for analysis:

- "One Laptop per Child at Home" Beuermann et al. (2015);
- "Changing climates of conflict: A social network experiment" Paluck et al. (2016);
- "Social Networks and the Decision to Insure" Cai et al. (2015).

Students may also implement the method using other empirical or simulated datasets that they find interesting.

References

- BEUERMANN, D. W., J. CRISTIA, S. CUETO, O. MALAMUD, AND Y. CRUZ-AGUAYO (2015): "One laptop per child at home: Short-term impacts from a randomized experiment in Peru," *American Economic Journal: Applied Economics*, 7, 53–80 (**).
- CAI, J., A. D. JANVRY, AND E. SADOULET (2015): "Social networks and the decision to insure," *American Economic Journal: Applied Economics*, 7, 81–108 (**).
- LEUNG, M. P. (2020): "Treatment and spillover effects under network interference," *Review of Economics and Statistics*, 102, 368–380 (**).
- PALUCK, E. L., H. SHEPHERD, AND P. M. ARONOW (2016): "Changing climates of conflict: A social network experiment in 56 schools," *Proceedings of the National Academy of Sciences*, 113, 566–571 (**).